Gregory J. Schwing, M.D., Ph.D.

Detroit, MI
gregory.schwing@med.wayne.edu · gregory-schwing-md-phd.github.io
Lab: openspineconsortium.com
Curriculum Vitae, October 2026

RESEARCH INTERESTS

Machine learning for medical imaging; computer vision for spine surgery and intraoperative fluoroscopy; curation and evaluation of large annotated imaging datasets; robustness of segmentation models to anatomical variation; edge deployment of clinical AI; generative deep learning for molecular design; GPU-accelerated Monte Carlo simulation; parallel graph algorithms and scientific software.

EDUCATION

2026 — **Ph.D., Computer Science, Wayne State University, Detroit, MI**
Dissertation: *Matchings and Molecules: Ball and Stick Models for Cheminformatics* (defended February 2024).
ProQuest 3355846225 · IPDPSW 2024 (CPU) · IPDPSW 2024 (GPU)
Advisor: Loren Schwiebert (Computer Science). Schwiebert–Potoff Laboratory (GOMC: GPU-Optimized Monte Carlo), co-PI Jeffrey Potoff (Chemical Engineering).

2026 — **M.D., Wayne State University School of Medicine, Detroit, MI**
Combined M.D./Ph.D. program, 2018–2026.

2018 — **B.S., Computer Science, University of New Orleans, New Orleans, LA**
Magna cum laude; Departmental Honors (thesis option). Senior honors thesis: *Implementation of Replica Exchange with Dynamic Scaling in GROMACS 2018*.
ScholarWorks UNO 117 · J Chem Inf Model 2021

2015 — **B.S., Biological Sciences, University of New Orleans, New Orleans, LA**
Magna cum laude.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

2026–present — **Founder and Research Director, OpenSpineConsortium (Schwing Lab)**
Wayne State University / Detroit Medical Center, Detroit, MI. Lead a research group of Wayne State medical students building open spine-imaging datasets and neural networks that name vertebral levels on intraoperative imaging to prevent wrong-level surgery.

2026–2027 — **Resident Physician (PGY-1), General Surgery**
Wayne State University / Detroit Medical Center, Detroit, MI. ACGME-accredited program.

2025–present — **Research Collaboration, Reisner Laboratory, Pediatric Neurosurgery (A. Reisner)**
Children's Healthcare of Atlanta / Emory University, Atlanta, GA. Liquid biopsy and diffuse midline glioma.

2018–2024 **Graduate Research Assistant, GPU-Optimized Monte Carlo (GOMC) Laboratory**
Wayne State University College of Engineering, Detroit, MI. Lead software developer of GOMC; the laboratory's only graduate assistant, supervised directly by both principal investigators, the chairs of Computer Science and Chemical Engineering.

Earlier research positions

2017–2018 **Summer Research Fellow, Computational Structural Biochemistry (C. Summa)**
University of New Orleans, New Orleans, LA.

2017 **Summer Research Fellow, Bioinformatics (C. Hicks)**
Louisiana State University Health Sciences Center, New Orleans, LA.

2016 **Clinical Research Associate, Cardiology (I. Basu-Ray)**
Tulane Hospital, New Orleans, LA.

2015 **Pediatric Oncology Education Fellow (C. Gawad)**
St. Jude Children's Research Hospital, Memphis, TN.

2014 **Undergraduate Research Assistant, Biochemistry (Liu Lab)**
University of New Orleans, New Orleans, LA.

HONORS AND AWARDS

2026 **Mentored project won first place, Detroit Annual Conference in Orthopaedics, Detroit, MI, October 2026 (as PI)**

2026 **Mentored projects won first and third place, oral presentations, Preston M. Hickey Medical Student Symposium, Wayne State University School of Medicine (as PI)**

2024 **Clerkship Honors, Obstetrics/Gynecology and Family Medicine**

2022 **Wayne State University Extramural Funding Incentive Award**

2022–2023 **Rumble Fellowship, Wayne State University (declined for the NIH F30)**

2018–2020 **Board of Governors Full Tuition Scholarship, Wayne State University School of Medicine (M.D./Ph.D. program)**

2018 **ACERS Scholarship (Achievement in Computer Science), University of New Orleans**

2018 **Departmental Honors in Computer Science; Magna cum laude, B.S. Computer Science**

2015 **Academic Excellence Award; Biological Sciences Achievement Award; Magna cum laude, B.S. Biological Sciences**

GRANTS AND FELLOWSHIPS

2022–2026 **NIH/NIGMS F30 Ruth L. Kirschstein Individual Predoctoral Fellowship, 1F30GM147909-01**
Myc Transcription Factor Inhibitor Design: Integrating Atomic and Mesoscale with Semi-Supervised Generative Deep Learning Models. Role: Principal Investigator (100% effort). Total direct costs: $133,350. NIH RePORTER.

Under review **AIDaS Seed Grant proposal, Wayne State University**
Role: Investigator (PI: Zhizhong Han; Co-PI: Loren Schwiebert). Submitted October 2026.

PUBLICATIONS

Names of medical-student mentees are marked with an asterisk (*).

Peer-reviewed conference papers

[1] **Schwing GJ**, Schwing P, Ismoilov M, Alnabahneh N, Schwiebert L. Not all spines are created equal: how CT segmenters fail on lumbosacral transitional vertebrae. *Conference on Neural Information Processing Systems (NeurIPS 2026), Evaluations and Datasets Track*. Accepted.

[2] **Schwing G**, Grosu D, Schwiebert L. Shared-memory parallel Edmonds blossom algorithm for maximum cardinality matching in general graphs. *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, pp. 530–539, 2024. doi:10.1109/IPDPSW63119.2024.00107

[3] **Schwing G**, Grosu D, Schwiebert L. Parallel maximum cardinality matching for general graphs on GPUs. *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, pp. 880–889, 2024. doi:10.1109/IPDPSW63119.2024.00157

Peer-reviewed journal articles

[1] **Schwing G**, Revuelta-Barbero JM, Beckman R, Reisner A. Peritoneal metastasis of histone 3 lysine 27-to-methionine-mutant diffuse midline glioma via ventriculoperitoneal shunt: laparoscopic confirmation of occult disease and the need for multimodal liquid biopsy. *BMJ Case Reports*, 2026. In press (accepted 23 October 2026).

[2] Schehr A*, Kim J*, **Schwing G**. OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research. *Cureus* 18(7):e112661, 2026. PMID 42598196. doi:10.7759/cureus.112661

[3] Barhaghi MS, Crawford B, **Schwing G**, Hardy DJ, Stone JE, Schwiebert L, Potoff J, Tajkhorshid E. Py-MCMD: Python software for performing hybrid Monte Carlo/molecular dynamics simulations with GOMC and NAMD. *Journal of Chemical Theory and Computation* 18(8):4983–4994, 2022. Journal cover. doi:10.1021/acs.jctc.1c00911

[4] Rick SW, **Schwing GJ**, Summa CM. An implementation of replica exchange with dynamical scaling for efficient large-scale simulations. *Journal of Chemical Information and Modeling* 61(2):810–818, 2021. doi:10.1021/acs.jcim.0c01236

[5] Nejahi Y, Barhaghi MS, **Schwing G**, Schwiebert L, Potoff J. Update 2.70 to "GOMC: GPU Optimized Monte Carlo for the simulation of phase equilibria and physical properties of complex fluids." *SoftwareX* 13:100627, 2021. doi:10.1016/j.softx.2020.100627

[6] Hoerger M, Wayser GR, **Schwing G**, Suzuki A, Perry LM. Impact of interdisciplinary outpatient specialty palliative care on survival and quality of life in adults with advanced cancer: a meta-analysis of randomized controlled trials. *Annals of Behavioral Medicine* 53(7):674–685, 2019. doi:10.1093/abm/kay077

[7] Basu-Ray I, Sudhakar D, **Schwing G**, Monlezun D, Zhang L, Shah SK, Pujara D, Ting K, Rafeh NA, Ali G, Cassidy M, Ellenbogen K, Levine G, Lam W, Mathuria N, Saeed M, Bunch J, Martin-Schild S, Gold M, Aryana A, Razavi M, Rasekh A. Complex left atrial appendage morphology is an independent risk factor for cryptogenic ischemic stroke. *Frontiers in Cardiovascular Medicine* 5:131, 2018. doi:10.3389/fcvm.2018.00131

Preprints and manuscripts under review

[1] **Schwing G**. SchwingNet: a convolutional-recurrent network for level naming in spine surgery, robust to transitional anatomy, versus registration-assisted neurosurgeons. Under review, *Radiology: Artificial Intelligence*, 2026.

[2] **Schwing G**, Schehr A*, Tekumulla A*, Khoushi M*, Christian R*, Hubers D*, Mahjoub F*, Saad H*, Sooch M*, Siddapureddy S*, McLellan M*, Kim J*, Ismoilov M, Alnabahneh N. CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy. Under review, *Medical Physics*, 2026. Preprint: arXiv:2609.22760

[3] Ayloo B, Kim J*, Shin H, **Schwing G**, Kantak P, Lee I. Concordance of frontier large language models (Claude Opus 4.7, ChatGPT 5-5, Gemini Pro 3.1) and EPIC-assigned codes in coding craniotomy procedures. Under review, *Journal of Neurosurgery*, 2026.

[4] **Schwing G**, Palese LL, Fernández A, Schwiebert L, Gatti DL. Molecular dynamics without molecules: searching the conformational space of proteins with generative neural networks. Preprint, arXiv:2206.04683, 2022.

Peer-reviewed abstracts

[1] **Schwing G**, Nagaraja T, Datta I, Lee I. Characterizing the transcriptomic recurrence signature of glioblastoma following laser interstitial thermal therapy. *Neurosurgery* (CNS Annual Meeting Supplement), 2026. In press.

[2] **Schwing G**. Not all spines are created equal: a single-shot CT segmentation model for preoperative detection of lumbar and thoracic end anomalies. *Clinical Neurosurgery* (CNS Annual Meeting Supplement), 2026. In press.

[3] **Schwing G**, Akhtar M*, Christian R*, Dafinescu S*, DesChamps B*, Fahmy J*, Hubers D*, Khoushi M*, Kim J*, Mahjoub F*, Saad H*, Schehr A*, Sooch M*, Vedagarbha N*, Weiss J*, Zaghloul R*. Beyond the cadaver lab: the OpenSpineConsortium as an asynchronous, open-source framework for medical student engagement in computational spine imaging research. *Clinical Neurosurgery* (CNS Annual Meeting Supplement), 2026. In press.

[4] **Schwing G**, Revuelta-Barbero JM, Beckman R, Reisner A. Peritoneal metastasis of histone 3 lysine 27-to-methionine-mutant diffuse midline glioma via ventriculoperitoneal shunt: laparoscopic confirmation of occult disease and the need for multimodal liquid biopsy. *Clinical Neurosurgery* (CNS Annual Meeting Supplement), 2026. In press.

[5] Ayloo B, Kim J*, Mitchell M, **Schwing G**, Shin H, Kantak P, Lee I. Concordance of frontier large language models (Claude Opus 4.7, ChatGPT 5-5, Gemini Pro 3.1) and EPIC coders in coding craniotomy procedures. *Clinical Neurosurgery* (CNS Annual Meeting Supplement), 2026. In press.

[6] Nagaraja TN, **Schwing GJ**, Datta I, Lee IY. Potential drug targets identified by characterizing the transcriptomic recurrence signatures of glioblastoma following laser interstitial thermal therapy. *Neuro-Oncology* (SNO Annual Meeting Supplement), 2026. In press.

[7] **Schwing G**, Moisi M, Hicks C. A novel bioinformatics approach to estimate hazard in pediatric high grade glioma. *Journal of Neurosurgery* 130 (AANS Annual Meeting Supplement), 2019.

PATENTS

[1] **Schwing G**. Automated vertebral level naming on intraoperative imaging (SchwingNet). Provisional patent application, Wayne State University Technology Commercialization, Tech ID 26-1901, submitted October 31, 2026.

THESES

[1] **Schwing GJ**. *Matchings and Molecules: Ball and Stick Models for Cheminformatics*. Ph.D. dissertation, Computer Science, Wayne State University, 2024. ProQuest 3355846225 · IPDPSW 2024 (CPU) · IPDPSW 2024 (GPU).

[2] **Schwing GJ**. *Implementation of Replica Exchange with Dynamic Scaling in GROMACS 2018*. Senior honors thesis, Computer Science, University of New Orleans, 2018. ScholarWorks UNO 117 · J Chem Inf Model 2021.

SOFTWARE AND DATASETS

- CTSpinoPelvic1K (2026): open CT dataset with spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy. arXiv:2609.22760; NeurIPS 2026 (accepted).
- OpenSpineConsortium (2026): open-source framework and summer workshop for medical-student research in computational spine imaging (openspineconsortium.com). *Cureus* 18(7):e112661, 2026.
- GOMC: GPU Optimized Monte Carlo (2018–2024): lead developer of the open-source GPU Monte Carlo engine for phase equilibria (gomc-wsu.org). *SoftwareX* 13:100627, 2021 (version 2.70).
- Py-MCMD (2022): lead programmer; hybrid Monte Carlo/molecular dynamics coupling of GOMC and NAMD, with collaborators at the University of Illinois Urbana-Champaign. *J Chem Theory Comput* 18(8):4983–4994, 2022 (journal cover).
- Parallel maximum cardinality matching (2024): shared-memory and GPU implementations of the Edmonds blossom algorithm for general graphs. IPDPSW 2024, pp. 530–539 and pp. 880–889.
- Replica exchange with dynamical scaling (2018): implementation in GROMACS 2018 (senior honors thesis, University of New Orleans). *J Chem Inf Model* 61(2):810–818, 2021.

PRESENTATIONS

Invited talks

[1] **Schwing G**. When the cord is compressed: neuroradiology essentials for surgical decision-making in degenerative spine disease. Invited talk, Michigan Association of Neurological Surgeons Annual Meeting, Crystal Mountain, MI, August 2026.

Contributed oral presentations

[1] **Schwing G**, Nagaraja T, Datta I, Lee I. Characterizing the transcriptomic recurrence signature of glioblastoma following laser interstitial thermal therapy. Congress of Neurological Surgeons Annual Meeting (Sunrise Session), Washington, DC, November 2026.

[2] **Schwing G**. Not all spines are created equal: a single-shot CT segmentation model for preoperative detection of lumbar and thoracic end anomalies. Congress of Neurological Surgeons Annual Meeting (Mini Podium), Washington, DC, November 2026.

[3] Nagaraja TN, **Schwing GJ**, Datta I, Lee IY. Potential drug targets identified by characterizing the transcriptomic recurrence signatures of glioblastoma following laser interstitial thermal therapy. Society for Neuro-Oncology Annual Meeting, Philadelphia, PA, November 2026.

[4] Schehr A*, Kim J*, **Schwing G**. Automated CT-based classification of iliac crest height relative to lumbosacral disc levels for LLIF planning. Detroit Annual Conference in Orthopaedics, Detroit, MI, October 2026. First place award.

[5] Schehr A*, Kim J*, Hubers D*, Saad H*, Fahmy J*, **Schwing G**. Automated CT-based Cobb angle measurement for scoliosis evaluation: a computational imaging pipeline across 1,156 cases. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026. First place, oral presentations.

[6] Zaghloul R*, Khoushi M*, Akhtar M*, Mahjoub F*, Schehr A*, **Schwing G**. Automated measurement of lumbar intervertebral disc heights from CT segmentation masks: a preliminary validation study. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026. Third place, oral presentations.

[7] **Schwing G**. Harnessing algorithmic disagreement for automated lumbosacral transitional vertebrae detection. 43rd Annual Meeting of the American Association of Clinical Anatomists (Tech Fair), Mayo Clinic, Rochester, MN, June 2026.

[8] **Schwing G**, Grosu D, Schwiebert L. Shared-memory parallel Edmonds blossom algorithm for maximum cardinality matching in general graphs. IEEE International Parallel and Distributed Processing Symposium Workshops, San Francisco, CA, May 2024.

[9] **Schwing G**, Grosu D, Schwiebert L. Parallel maximum cardinality matching for general graphs on GPUs. IEEE International Parallel and Distributed Processing Symposium Workshops, San Francisco, CA, May 2024.

[10] **Schwing G**, Barhaghi MS, Crawford B, Schwiebert L, Potoff J. Leveraging the Wolf method for electrostatics to extend time and length scales accessible by Monte Carlo simulations. AIChE Annual Meeting, Phoenix, AZ, November 2022.

[11] Crawford B, Barhaghi MS, **Schwing G**, Hardy D, Stone J, Schwiebert L, Potoff J, Tajkhorshid E. Py-MCMD: hybrid Monte Carlo–molecular dynamics simulations. AIChE Annual Meeting, Phoenix, AZ, November 2022.

[12] Crawford B, Barhaghi MS, **Schwing G**, Hardy D, Stone J, Schwiebert L, Potoff J, Tajkhorshid E. Hybrid Monte Carlo–molecular dynamics simulations. AIChE Annual Meeting (virtual), November 2021.

[13] Nejahi Y, **Schwing G**, Schwiebert L, Potoff J. Unsupervised optimization of non-bonded parameters in molecular mechanics force fields using the particle swarm method. AIChE Annual Meeting (virtual), November 2020.

[14] **Schwing G**, Moisi M, Hicks C. A novel bioinformatics approach to estimate hazard in pediatric high grade glioma. American Association of Neurological Surgeons Annual Scientific Meeting, San Diego, CA, April 2019.

Posters

[1] **Schwing G**, Akhtar M*, Christian R*, Dafinescu S*, DesChamps B*, Fahmy J*, Hubers D*, Khoushi M*, Kim J*, Mahjoub F*, Saad H*, Schehr A*, Sooch M*, Vedagarbha N*, Weiss J*, Zaghloul R*. Beyond the cadaver lab: the OpenSpineConsortium as an asynchronous, open-source framework for medical student engagement in computational spine imaging research. Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[2] Ayloo B, Kim J*, Mitchell M, **Schwing G**, Shin H, Kantak P, Lee I. Concordance of frontier large language models (Claude Opus 4.7, ChatGPT 5-5, Gemini Pro 3.1) and EPIC coders in coding craniotomy procedures. Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[3] **Schwing G**, Revuelta-Barbero JM, Beckman R, Reisner A. Peritoneal metastasis of histone 3 lysine 27-to-methionine-mutant diffuse midline glioma via ventriculoperitoneal shunt: laparoscopic confirmation of occult disease and the need for multimodal liquid biopsy. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[4] Saad H*, Fahmy J*, DesChamps B*, Schehr A*, Hubers D*, Kim J*, **Schwing G**. Sagittal lumbar lordosis quantification via automated CT segmentation analysis: a TRALL angle computational pipeline. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[5] Vedagarbha N*, Sooch M*, Thomas L*, **Schwing G**. Opportunistic screening for vertebral degeneration using automated analysis of vertebral Hounsfield units. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[6] Dafinescu S*, Schehr A*, **Schwing G**. Automated 3D vertebral centroid trajectory analysis from CT segmentation masks. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[7] Zaghloul R*, Zeidan D*, Zvinys L*, **Schwing G**. Automated measurement of lumbar intervertebral disc heights from CT segmentation masks: a preliminary validation study. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[8] Hubers D*, Schehr A*, Kim J*, Fahmy J*, Saad H*, **Schwing G**. Automated lumbar spinal stenosis screening via CT morphometry: normative Torg ratio values across L1–L5 in a multi-configuration dataset. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[9] Schehr A*, Kim J*, Hubers D*, Saad H*, Fahmy J*, **Schwing G**. Enhancing clinical workflow in scoliosis evaluation: automated CT-derived Cobb angle measurement using a geometry-based pipeline. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[10] Weiss J*, Yazhbin A*, Schehr A*, **Schwing G**. Automated CT-based extraction of pelvic incidence and sacral slope for spinopelvic risk stratification. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[11] Christian R*, Schehr A*, **Schwing G**. Automated vertebral wedging index from CT: a geometry-based biomarker for lumbar compression deformity. Digital poster, Congress of Neurological Surgeons Annual Meeting, Washington, DC, November 2026.

[12] Saad H*, Fahmy J*, DesChamps B*, Schehr A*, Hubers D*, Kim J*, **Schwing G**. Automated TRALL angle computation from CT segmentation: a scalable pipeline for lumbar lordosis quantification. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026.

[13] Kim J*, Schehr A*, Hubers D*, Fahmy J*, Saad H*, **Schwing G**. Straightening out the curve: automated analysis of lordosis distribution index measurement using CT segmentation masks. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026.

[14] Hubers D*, Schehr A*, Kim J*, Fahmy J*, Saad H*, **Schwing G**. CT-based automated morphometric analysis of the lumbar spinal canal: normative Torg ratio and canal diameter values across L1–L5. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026.

[15] Fahmy J*, Saad H*, Schehr A*, **Schwing G**. Automated quantification of lumbar spondylolisthesis using CT-derived vertebral landmarks. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026.

[16] Vedagarbha N*, Sooch M*, Thomas L*, **Schwing G**. Opportunistic screening for vertebral degeneration using automated analysis of vertebral Hounsfield units. Preston M. Hickey Medical Student Symposium, Detroit, MI, August 2026.

[17] **Schwing G**. A combinatorial graph algorithm implemented on the GPU and parameter optimization using Gaussian processes. Graduate Research Symposium, Wayne State University, Detroit, MI, February 2024.

[18] Nejahi Y, Barhaghi MS, **Schwing G**, Schwiebert L, Potoff J. GPU Optimized Monte Carlo version 2.50. AIChE Annual Meeting (virtual), November 2020.

[19] **Schwing G**, Hicks C. A novel bioinformatics approach for mapping the genomic landscape of pediatric glioblastoma multiforme. 8th Annual Southeastern Medical Scientist Symposium, Atlanta, GA, November 2017.

[20] Ray IB, **Schwing G**, Middour T, Monlezun DJ, Katikitala K, Koranne K, Martin-Schild S, Lam W, Afsar H, Giorgberidze I, Cheng J, Saeed M. Heart failure with preserved ejection fraction is associated with cardioembolic stroke independent of history of atrial fibrillation. European Society of Cardiology Congress, Barcelona, Spain, August 2017.

[21] **Schwing GJ**, Hicks C. A novel integrative bioinformatics approach for mapping the genomic landscape of pediatric glioblastoma multiforme. Summer Research Internship Program Poster Session, Louisiana State University Health Sciences Center, New Orleans, LA, July 2017.

[22] Ray IB, **Schwing G**, Middour T, Monlezun DJ, Katikitala K, Koranne K, Martin-Schild S, Lam W, Afsar H, Giorgberidze I, Cheng J, Saeed M. Significant diastolic dysfunction is associated with cardioembolic stroke independent of history of atrial fibrillation. 74th World Congress on Acute Heart Failure, Paris, France, April 2017.

[23] Ray IB, **Schwing G**, Middour T, Monlezun DJ, Katikitala K, Koranne K, Cassidy M, Martin-Schild S, Lam W, Afsar H, Giorgberidze I, Cheng J, Saeed M. Significant diastolic dysfunction is associated with cardioembolic stroke independent of history of atrial fibrillation. American College of Cardiology Annual Scientific Session, Washington, DC, March 2017.

[24] Ray IB, **Schwing G**, Novak J, Monlezun DJ, Boja H, Deere B, Duque J, Park M, Lawrence J, Middour T, Morris S, Boulad A, Carry B, Abi Rafey N, Ali G, Anwar A, LeJemtel T, Martin-Schild S, Cassidy M. Complex left atrial appendage morphology is an independent risk factor for cryptogenic stroke. Atrial Fibrillation 22nd Annual Symposium, Orlando, FL, January 2017.

[25] Ray IB, **Schwing G**, Novak J, Monlezun DJ, Boja H, Deere B, Duque J, Park M, Lawrence J, Middour T, Morris S, Boulad A, Carry B, Abi Rafey N, Ali G, Anwar A, LeJemtel T, Martin-Schild S, Cassidy M. Complex left atrial appendage morphology increases risk of cryptogenic stroke. American Heart Association Scientific Sessions, New Orleans, LA, November 2016.

TEACHING AND MENTORING

2026	**Director, OpenSpineConsortium Summer Workshop (MSK Radiology and AI)** First- and second-year Wayne State medical students (openspineconsortium.com/#workshop). Educational outcomes published in *Cureus* 18(7):e112661, 2026.
2026–present	**Research Mentor, OpenSpineConsortium** Mentor to 22 Wayne State medical-student researchers (every co-author marked with an asterisk above) in dataset annotation, computational imaging and manuscript writing; mentees won first and third place at the Preston M. Hickey Medical Student Symposium and have posters accepted at CNS 2026 (November).
2026–present	**Clinical Teaching, General Surgery Internship** Third- and fourth-year medical students and physician assistant students, as a general surgery intern.
2019–2020	**Case-Based Learning Facilitator, Wayne State University School of Medicine** First-year medical student small groups, including the ALS case series.
2018	**Facilitator, In Silico Computational Biology Seminar** Graduate students.

SERVICE AND OUTREACH

2026	**Organizer, "Celebrate Life" event for pediatric hematology/oncology patients, with the Detroit Football Club's Trunk or Treat**
2024–2026	**President, Catholic Medical Student Association, Wayne State University School of Medicine**
2018, 2024	**Volunteer and M1/M2 supervisor, Robert R. Frank Student Run Free Clinic, Detroit, MI**
2018–2019	**Literacy Teacher, Auntie Na's Village, Detroit, MI**
2016–2019	**Child Life Volunteer, Children's Hospital of Michigan (2019) and Children's Hospital New Orleans (2016–2017)**
2015	**Happy Cart Volunteer, St. Jude Children's Research Hospital, Memphis, TN**
2011–2012	**Tutor, Mobile, AL public schools (middle-school students)**

PROFESSIONAL MEMBERSHIPS

Association for Computing Machinery (2024–present); American Institute of Chemical Engineers (2024–present); Congress of Neurological Surgeons (2024–present); American Medical Association (2024–present); American Association of Neurological Surgeons (2019–present).

LICENSURE AND CERTIFICATION

2026–2027	**Michigan Medical Doctor Educational Limited License** Expires June 2027; full and unrestricted Michigan license expected July 2027. NPI 1033907571.
	USMLE Step 1 and Step 2 CK (passed)
2026–2028	**ACLS Provider, American Heart Association** Issued May 2026, valid through May 2028.
2026–2028	**BLS Provider, American Heart Association** Issued April 2026, valid through April 2028.
2024	**CITI Program: Human Subjects Research; Health Information Privacy and Security (HIPS)**

TECHNICAL SKILLS

- Languages: Python, C++, CUDA, SQL, SAS.
- Machine learning: PyTorch, nnU-Net, MONAI; convolutional, recurrent and generative neural networks; medical image segmentation.
- Scientific computing: Monte Carlo and molecular dynamics simulation (GOMC, GROMACS, NAMD; hybrid MC/MD methods); GPU and shared-memory parallel programming; Slurm-managed HPC clusters; Gaussian-process and particle-swarm parameter optimization.
- Bioinformatics: RNA-seq differential expression analysis; Ingenuity Pathway Analysis.
- Imaging: CT, radiographs, fluoroscopy, DICOM. Tools: Git, Linux.

REFERENCES

Available on request.